

Saadbin Khan

Website: saadbin.github.io | Email: saadbin@vt.edu | LinkedIn | Google Scholar | Blacksburg, VA | (540)-449-1156

HIGHLIGHTS

- ◇ 9+ years of experience in CFD simulations, multiscale heat/mass transport modeling, and particle-based DSMC simulations.
- ◇ 5+ years of mechanical design experience with SolidWorks CAD and high-fidelity custom meshing.
- ◇ Experience with data-driven inverse modeling, reduced-order modeling, and high-performance computing.
- ◇ 5 publications, 1 book chapter, 10+ conference/symposium/poster presentations, and 2 journal reviews. (Full list: [📄](#))

EDUCATION

- ◇ **Ph.D. in Engineering Mechanics** 2021 - 2026 (exp.)
Virginia Tech (CGPA: 3.89) Blacksburg, VA
- ◇ **M.S. in Mechanical and Aerospace Engineering** 2017 - 2019
Oklahoma State University (CGPA: 3.63) (Thesis: [📄](#)) Stillwater, OK
- ◇ **B.Sc. in Mechanical Engineering** 2011 - 2016
Bangladesh University of Engineering and Technology (Thesis: [📄](#)) Dhaka, Bangladesh

TECHNICAL STRENGTHS

- Programming:** Python, Fortran, MATLAB, C++, C, Mathematica
- Design & Simulation:** SolidWorks, OpenFOAM, ANSYS Fluent, COMSOL, DSMC, OpenLB, Open MPI
- Database & Post-processing:** ParaView, Tecplot, Tracker, ImageJ, Meshmixer, Microsoft Excel

NOTABLE PROJECTS [\[SEE FULL LIST HERE: 📄\]](#)

- ◇ **The role of hydrodynamic slip and fine tracheal geometry in insect respiratory flow** [Portfolio: [📄](#)] 2022 - 2026
 - Funder: NSF | Tools: OpenFOAM, C++, Python, SolidWorks, DSMC, OpenLB, Paraview
 - Outcome: Demonstrated that slip and fine wall structures enhance tracheal flow beyond classical no-slip predictions; consequently developed a microscale-physics informed modified Murray's law applicable to microfluidic devices.
- ◇ **DNS of wavy channel flow to understand the near-wall turbulence structure** [Portfolio: [📄](#)] 2017 - 2019
 - Funder: NASA Oklahoma EPSCoR | Tools: Fortran, Python, Open MPI, ANSYS Fluent
 - Outcome: Simulated 3D turbulent flow at $Re \approx 2800$ over sinusoidal roughness using MPI parallelization with HPC, showing significantly stronger Reynolds stress distributions and turbulence structures relative to flat-wall channel flow.
- ◇ **Improved atmospheric plume source inversion model for remote leak detection** [Portfolio: [📄](#)] 2018
 - Funder: Baker Hughes General Electric | Tools: Python
 - Outcome: Achieved 99.2% accurate source localization using 1% concentration measurement data in 2D domain.
- ◇ **Mixed convection heat transfer in nanofluids with MHD effects in a porous medium** [Portfolio: [📄](#)] 2016
 - Tools: COMSOL, Tecplot
 - Outcome: Identified optimal heat transfer performance at 15% waviness to cavity width ratio and 9% CNT-nanoparticle concentration, with consistent increase in average Nusselt number across operating conditions

RELEVANT EXPERIENCE

- ◇ **Research and Teaching Assistant** 2021 - Present
Dept. of Mechanical Engineering, Dept. of Biomedical Engineering, & BIOTRANS, Virginia Tech Blacksburg, VA
- ◇ **Instructor of Record for Thermodynamics (ME-2134)** 2025
Dept. of Mechanical Engineering, Virginia Tech Blacksburg, VA
- ◇ **Research and Teaching Assistant** 2017 - 2019
Dept. of Mechanical & Aerospace Engineering, Oklahoma State University Stillwater, OK
- ◇ **Engineering Intern** 2016
Bangladesh Atomic Energy Commission Dhaka, Bangladesh
- ◇ **Other Services**
 - Journal Reviewer (MDPI, Elsevier); Webmaster (GEMS, Virginia Tech); Student Administrative Assistant (BIOTRANS, Virginia Tech); Public Outreach Manager (Bangladeshi Students Association, Oklahoma State University).

AWARDS

- **Alice & Dan Pletta Scholarship**, Engineering Mechanics Graduate Program, Virginia Tech 2026
- **BIOTRANS Fellowship**, Interdisciplinary Graduate Education Programs, Virginia Tech 2021 - 2022
- **Champion**, 6th and 8th Bangladesh Mathematical Olympiads 2008 & 2010